DGMS 2019 • SECTION L: ENVIRONMENTAL & ENERGY GEOLOGY

Question 86: Geothermal Energy Systems: Plate Boundaries & Thermal Gradients

Official Examination Guide • Hydrothermal Convection, Depth of Thermal Penetration & Hyperthermal Gradients

OFFICIAL EXAMINATION QUESTION

Consider the following statements regarding geothermal energy:

1. Most areas where this energy is being tapped are along or near the plate boundaries.

2. Seasonal variation in the earth's surface temperature significantly affects this energy.

3. The places on the earth where groundwater circulates near magma are the easier sites of extraction of this energy.

4. The places on the earth having geothermal gradient around 25–50°C/km are considered as good sites.

Which of the statements given above is/are correct?

(a) 1 and 3 only CORRECT ANSWER

(b) 2 and 4 only

(c) 1, 3 and 4 only

(d) 1, 2, 3 and 4

1. Executive Summary & Core Principle

Geothermal energy utilizes the natural heat of the Earth's interior. In global geophysics (Fowler, *The Solid Earth*; Turcotte & Schubert, *Geodynamics*):

- **Statement 1 is TRUE:** The vast majority of commercially exploited high-enthalpy geothermal fields globally are concentrated along active tectonic **plate boundaries** (convergent subduction volcanic arcs, continental rifts, and mid-ocean ridges). Famous examples include the Pacific "Ring of Fire" (The Geysers, Salton Sea in California; Cerro Prieto in Mexico; the Philippines; Indonesia; New Zealand), Iceland along the Mid-Atlantic Ridge, and the East African Rift.
- **Statement 2 is FALSE:** Atmospheric seasonal temperature variations penetrate into the crust only to shallow depths of roughly **10 to 15 meters**. Below this shallow thermal attenuation skin depth, ground temperatures remain completely constant year-round. Commercial geothermal extraction taps deep reservoirs at depths of **1,000 to 3,500 meters (1–3.5 km)**, which are entirely unaffected by surface seasonal temperature changes!
- **Statement 3 is TRUE:** The most easily exploitable geothermal fields are natural **hydrothermal convection systems**. Deeply circulating meteoric groundwater encounters shallow magmatic bodies or hot intrusive plutons, heats to 150°C–350°C, and rises through permeable fractures to form pressurized steam or hot water that can be tapped with conventional production wells.
- **Statement 4 is FALSE:** A geothermal gradient of **25°C to 30°C/km** (or 25–50°C/km) is the *average background continental geothermal gradient*! A gradient of 30°C/km is far too low for economic hydrothermal power generation (it would require drilling un-economically deep to reach usable steam temperatures). Commercially viable geothermal fields require **anomalously high hyperthermal gradients (> 100°C/km to 250°C/km)**.

Therefore, only Statements **1 and 3** are correct, making Option **(a) 1 and 3 only** the correct answer.

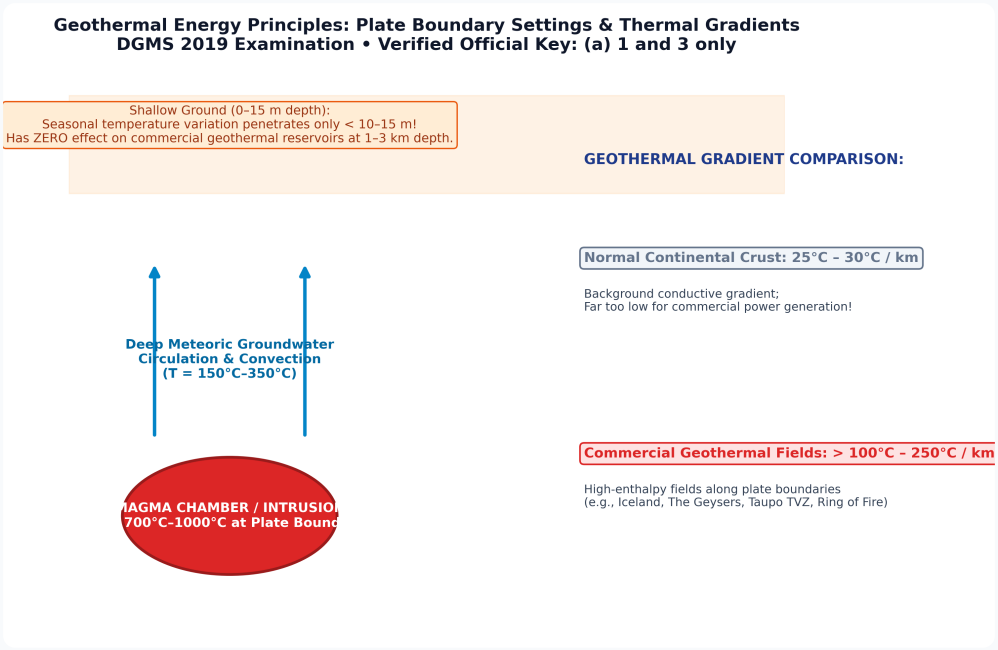


Figure 86.1: Hydrothermal convective circulation above magma chambers at plate boundaries contrasted with normal crustal geothermal gradients and shallow seasonal thermal skin depths (Grounded in Fowler Solid Earth and Turcotte & Schubert).

2. Geothermal Classification: Normal vs. Hyperthermal Gradients

Geological Environment	Geothermal Gradient	Heat Flow (q)	Geothermal Potential & Suitability
Normal Continental Cratons	**25°C – 30°C / km**	$40 – 60\text{ mW/m}^2$	**Unfavorable for electricity.** Only suitable for low-temperature ground-source heat pumps at shallow depths.
Sedimentary Basin Warm Aquifers	$30^\circ\text{C} – 50^\circ\text{C} / \text{km}$	$60 – 80\text{ mW/m}^2$	Low-to-medium enthalpy; suitable only for direct district heating or binary

Geological Environment	Geothermal Gradient	Heat Flow (q)	Geothermal Potential & Suitability
			Organic Rankine Cycle (ORC) plants.
Active Plate Margins & Magmatic Zones	**> 100°C – 300°C / km**	**> 150 – 500 mW/m²**	**IDEAL / COMMERCIAL.** High-enthalpy dry steam and flash-steam power generation (e.g., Puga Valley in Ladakh, Manikaran in HP, India; The Geysers, USA).

3. Evaluation of Exam Statements

- **Statement 1: CORRECT.** Plate boundary volcanism provides shallow crustal magma chambers necessary to heat surrounding groundwater.
- **Statement 2: INCORRECT.** Seasonal surface temperature variations attenuate to zero below 15 m; commercial reservoirs are at 1–3 km depth.
- **Statement 3: CORRECT.** Magma-driven hydrothermal systems provide natural high-permeability, high-pressure steam extraction sites.
- **Statement 4: INCORRECT.** 25–50°C/km represents typical background continental conditions, not the high-gradient hyperthermal anomaly required for geothermal power.

> **Key Takeaway:** High geothermal energy relies on plate margins and shallow magma convection (>100°C/km). Seasonal surface changes have zero effect at depth. Hence, **(a) 1 and 3 only** is the correct answer.

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